

Designed and assembled a fully articulated Robotic Leg mechanism in SolidWorks from scratch — using a crank-driven linkage system with a hip joint, knee joint, and flat foot.

- **Mates & Constraints** — Applied coincident, concentric, and hinge mates to bring individual parts together as a working mechanism.
- **Motion Study** — Simulated real movement across two motion studies to test and validate how each joint interacts under motion.
- **Assembly Workflow** — Managed multiple components, sub-assemblies, and configurations throughout the build.
- **Appearance & Rendering** — Added materials and used PhotoView 360 to produce a polished final presentation.

Skills Applied: 3D Modeling · Advanced Assembly Mates · Kinematics · Interference Detection · Motion Study